

Math 242, S19
Exam 1

Name: *Answer*

Instructions:

- There are totally 5 problems.
- You must show all of your work to receive a credit for a correct answer.

Problem	Points	Score
1	20	
2	22	
3	22	
4	20	
5	16	
Total	100	

Good Luck !

Problem 1. (20 points) Find the particular solution of the initial value problems:

(a) $\frac{dy}{dx} = \frac{-3}{\sqrt{2x+1}}, \quad y(4) = -1.$

$$y = \int \frac{-3}{\sqrt{2x+1}} dx = -3\sqrt{2x+1} + C$$

$$y(4) = -1 \Rightarrow -3\sqrt{2 \cdot 4 + 1} + C = -1$$

$$-3 \cdot 3 + C = -1$$

$$C = -1 + 9 = 8$$

$$\Rightarrow \underline{y(x) = -3\sqrt{2x+1} + 8}$$

(b) $(1+x^2)\frac{dy}{dx} - 2x(1+x^2)y = 2xe^{x^2}, \quad y(0) = 2.$

$$\frac{dy}{dx} - 2xy = \frac{2xe^{x^2}}{1+x^2} \leftarrow \text{standard form}$$

$$P(x) = -2x, \quad Q(x) = \frac{2xe^{x^2}}{1+x^2}$$

$$\rightarrow \text{Integrating factor } f(x) = e^{\int P(x)dx} = e^{\int -2x dx} = e^{-x^2}$$

$$\text{Thus } \frac{d}{dx}[y e^{-x^2}] = \frac{2x e^{x^2}}{1+x^2} \cdot e^{-x^2}$$

$$y e^{-x^2} = \int \frac{2x}{1+x^2} dx$$

$$y e^{-x^2} = \ln(1+x^2) + C$$

$$y = e^{x^2}(\ln(1+x^2) + C)$$

$$y(0) = 2 \Rightarrow e^0(\ln 1 + C) = 2 \Rightarrow C = 2$$

$$\Rightarrow \underline{y(x) = e^{x^2}(\ln(1+x^2) + 2)}$$

Problem 2. (22 points) Find a general solution for the following differential equations.

(a) $\frac{dy}{dx} = 1 + x^2 - y - x^2y.$ (Hint: Factor the right-hand side.)

$$\frac{dy}{dx} = (1+x^2)(1-y)$$

$$\frac{1}{1-y} dy = (1+x^2) dx$$

$$\int \frac{1}{1-y} dy = \int (1+x^2) dx$$

$$-\ln|1-y| = x + \frac{x^3}{3} + C$$

$$|1-y| = e^{-x - \frac{x^3}{3} + C}$$

$$1-y = Ae^{-x - \frac{x^3}{3}}$$

$$\underline{y = 1 - Ae^{-x - \frac{x^3}{3}}}$$

(b) $(3x^2 + 2y^2)dx + (4xy + 6y^2)dy = 0$

$$M(x,y) = 3x^2 + 2y^2 \Rightarrow \frac{\partial M}{\partial y} = 4y \Rightarrow \text{exact!}$$

$$N(x,y) = 4xy + 6y^2 \Rightarrow \frac{\partial N}{\partial x} = 4y$$

$$\begin{aligned} \text{Thus } F(x,y) &= \int M(x,y) dx = \int (3x^2 + 2y^2) dx \\ &= x^3 + 2xy^2 + g(y) \end{aligned}$$

$$\begin{aligned} \frac{\partial F}{\partial y} &= N \Rightarrow 4xy + g'(y) = 4xy + 6y^2 \\ g'(y) &= 6y^2 \Rightarrow g(y) = 2y^3 + C \end{aligned}$$

$$\text{Thus } F(x,y) = x^3 + 2xy^2 + 2y^3 + C$$

and the solution is given by

$$\underline{x^3 + 2xy^2 + 2y^3 = C}$$

Problem 3. (22 points) Find a general solution for the following differential equations:

(a) $\frac{dy}{dx} = (x + 4y)^2$

let $V = x + 4y$, then $y = \frac{1}{4}V - \frac{1}{4}x \Rightarrow \frac{dy}{dx} = \frac{1}{4}\frac{dV}{dx} - \frac{1}{4}$
 thus $\frac{1}{4}\frac{dV}{dx} - \frac{1}{4} = V^2$

$$\frac{dV}{dx} = 1 + 4V^2$$

$$\frac{1}{1+4V^2}dV = dx$$

$$\int \frac{1}{1+4V^2}dV = \int dx$$

$$\frac{1}{2}\arctan(2V) = x + C_1$$

$$\arctan(2V) = 2x + C$$

$$\Rightarrow V = \frac{1}{2}\tan(2x + C)$$

thus $x + 4y = \frac{1}{2}\tan(2x + C)$

$$y = \frac{1}{4}\left(\frac{1}{2}\tan(2x + C) - x\right)$$

$$\Rightarrow \underline{y(x) = \frac{1}{8}\tan(2x + C) - \frac{1}{4}x}$$

(b) $2xyy' = x^2 + 2y^2$

$$y' = \frac{x^2 + 2y^2}{2xy} = \frac{1}{2}\left(\frac{x}{y}\right) + \frac{y}{x}$$

let $V = \frac{y}{x} \Rightarrow y = Vx$ and $dy = x\frac{dV}{dx} + V$

thus $x\frac{dV}{dx} + V = \frac{1}{2}\frac{1}{V} + V$

$$x\frac{dV}{dx} = \frac{1}{2V} \Rightarrow 2VdV = \frac{1}{x}dx$$

$$\int 2VdV = \int \frac{1}{x}dx$$

$$V^2 = \ln|x| + C$$

$$\Rightarrow V = \pm\sqrt{\ln|x| + C}$$

$$\Rightarrow \frac{y}{x} = \pm\sqrt{\ln|x| + C}$$

$$\Rightarrow \underline{y(x) = x\sqrt{\ln|x| + C} \text{ or } y(x) = -x\sqrt{\ln|x| + C}}$$

Problem 4. (20 points) A certain city has a population of 25000 initially and a population of 30000 after 10 years. Assume that its population grows at a constant rate k per year.

(a) Write down the differential equation for the population of the city $P(t)$ and find its general solution.

$$\begin{aligned}\frac{dP}{dt} &= kP \\ \int \frac{1}{P} dP &= \int k dt \\ \ln P &= kt + C \quad (P > 0) \\ P &= e^{kt+C} \\ \Rightarrow P(t) &= A e^{kt}\end{aligned}$$

(b) Determine all constant parameters in the solution using the given conditions.

$$P(0) = 25000 \Rightarrow A e^0 = 25000 \Rightarrow A = 25000$$

$$\Rightarrow P(t) = 25000 e^{kt}$$

$$P(10) = 30000 \Rightarrow 25000 e^{10k} = 30000$$

$$e^{10k} = \frac{30000}{25000} = \frac{6}{5}$$

$$\Rightarrow 10k = \ln\left(\frac{6}{5}\right)$$

$$\Rightarrow k = \frac{1}{10} \ln\left(\frac{6}{5}\right)$$

$$\Rightarrow P(t) = 25000 e^{\frac{t}{10} \ln\left(\frac{6}{5}\right)}$$

(c) How long does it take for the population to reach 40000?

$$P(t) = 40000 \Rightarrow 25000 e^{\frac{t}{10} \ln\left(\frac{6}{5}\right)} = 40000$$

$$e^{\frac{t}{10} \ln\left(\frac{6}{5}\right)} = \frac{8}{5}$$

$$\frac{t}{10} \ln\left(\frac{6}{5}\right) = \ln\left(\frac{8}{5}\right)$$

$$t = 10 \cdot \frac{\ln\left(\frac{8}{5}\right)}{\ln\left(\frac{6}{5}\right)} \approx 25.78 \text{ years}$$

Problem 5. (16 points) Given the following initial value problem:

$$\frac{dy}{dt} = 3y^2 - 18y, \quad y(0) = 5.$$

(a) Find all critical points of the given autonomous differential equation and then determine whether it is stable or unstable for each of them.

$$f(y) = 3y^2 - 18y = 3y(y-6)$$

$$3y(y-6) = 0 \Rightarrow \underline{y=0}, \underline{y=6} \rightarrow \text{two critical points}$$

$y' > 0 \quad y' < 0 \quad y' > 0$
 $\rightarrow \quad \leftarrow \quad \rightarrow$
 $0 \qquad 6$

$y=0 \rightarrow \text{stable}$
 $y=6 \rightarrow \text{unstable}$

(b) Solve the given initial value problem for $y(t)$ and find $\lim_{t \rightarrow +\infty} y(t)$.

$$\frac{dy}{dt} = 3y(y-6)$$

$$\int \frac{1}{y(y-6)} = \int 3 dt$$

$$\int \frac{1}{6} \left(\frac{1}{y-6} - \frac{1}{y} \right) = 3t + C_1$$

$$\ln|y-6| - \ln|y| = 18t + C$$

$$\ln \left| \frac{y-6}{y} \right| = 18t + C$$

$$\left| \frac{y-6}{y} \right| = e^{18t+C}$$

$$\frac{y-6}{y} = A e^{18t}$$

$$y(0) = 5 \Rightarrow \frac{5-6}{5} = A$$

$$A = -\frac{1}{5}$$

$$\Rightarrow \frac{y-6}{y} = -\frac{1}{5} e^{18t}$$

$1 - \frac{6}{y} = -\frac{1}{5} e^{18t}$
 $\frac{6}{y} = \frac{1}{5} e^{18t} + 1$
 $y = \frac{6}{1 + \frac{1}{5} e^{18t}} \quad \text{or} \quad \frac{30}{5 + e^{18t}}$

$$\lim_{t \rightarrow +\infty} y(t) = \lim_{t \rightarrow +\infty} \frac{6}{1 + \frac{1}{5} e^{18t}} = 0$$